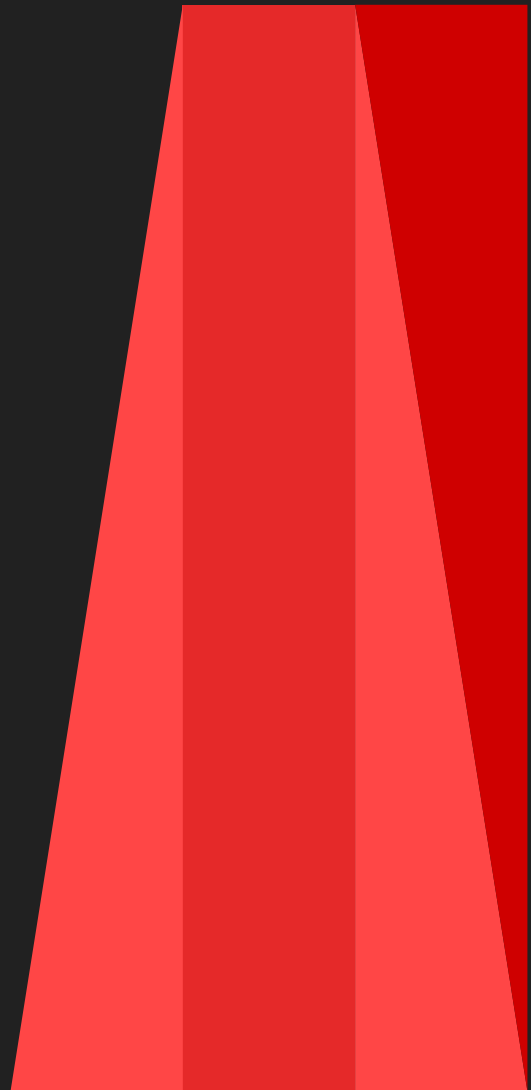


Nikhil Trivedi

Gesture Controlled
Differential Robotic Arm

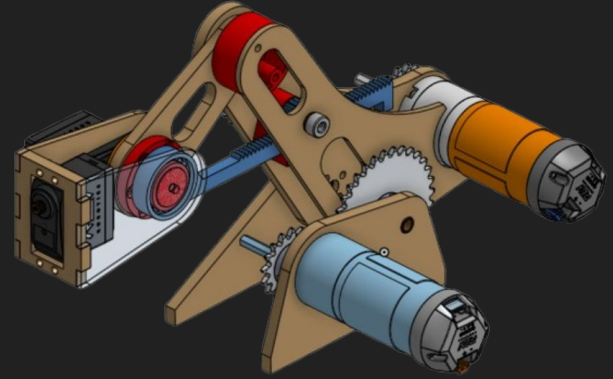


The goal of this independent project was to design and build a multi-joint robotic arm using FTC components that could be controlled both through traditional joystick input and later through hand gesture recognition. I initially planned a simple 3-axis robotic arm, but the design evolved into a more advanced differential joint system to improve strength, smoothness, and range of motion.

Planning & Design

I began by designing a basic robotic arm concept with multiple degrees of freedom. Early planning focused on achieving stable joint rotation and predictable motion using FTC-compatible motors and servos.

The design then evolved into a differential joint architecture, where motion is distributed across gears to allow smoother and more stable control compared to a single-motor joint.



Early 3-axis robotic arm concept

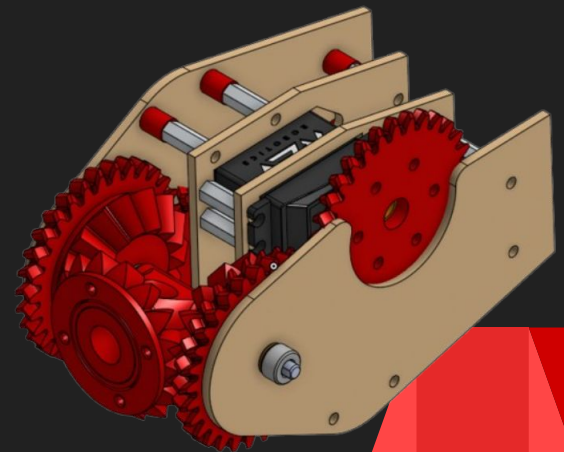
Key design decisions included:

- Using a differential mechanism instead of simple hinge joints for smoother motion control
- Implementing herringbone gears to improve alignment and reduce backlash under load
- Using bevel gears to transfer motion efficiently within the differential joint system
- Structuring the arm in modular segments so each joint could be tested independently before full assembly

Differential Joint Prototype

To validate the differential arm concept, I first built a single differential joint using REV Smart Servos. This prototype allowed me to test the mechanical design before committing to a full multi-joint arm.

- Herringbone gears for improved alignment and strength
- Bevel gears to create the differential mechanism
- A ball bearing to support smooth rotation under load



Differential Joint

Initial Control System

To test the prototype, I developed a simple TeleOp program using D-pad controls.

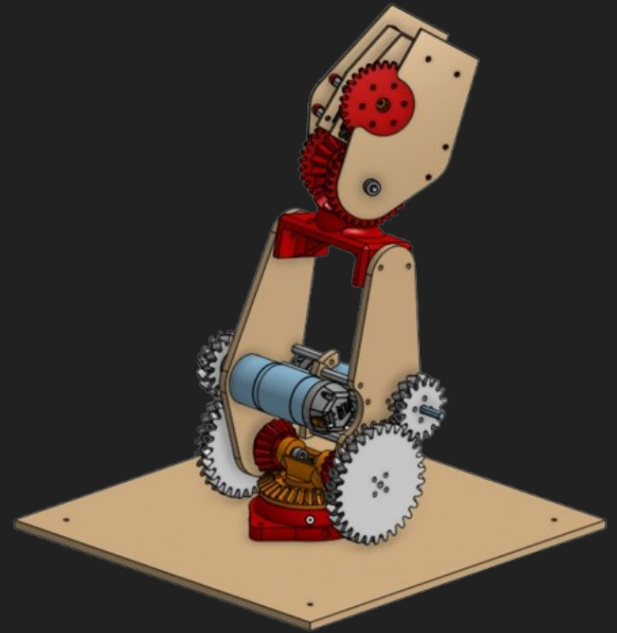
- Moved the joint in discrete steps
- Allowed controlled testing of range of motion
- Verified that the mechanical system responded consistently to commands

This stage focused on validating the differential mechanism before developing more advanced controls.

Expanding to a Multi-Joint Arm

After the first joint proved successful, I constructed a second, larger differential joint using REV HD Hex 40:1 motors. This became the base joint of the arm, while the original servo-powered joint became the upper joint.

The larger joint was designed to support greater loads and provide the strength needed for a full robotic arm structure.



Robotic Arm Final CAD model

Advanced Joystick Control

With the mechanical system functioning reliably, I developed a more advanced TeleOp control system.

- Joint position continuously followed joystick movement
- Movement was limited by joystick boundaries
- Returning the joystick to center returned the arm toward its neutral position

This created a more intuitive and responsive control system, allowing for precise positioning.

```
double tilt = 0.5 + (-gamepad1.left_stick_y * 0.25);  
double maxSpin = Math.min(1.0 - tilt, tilt);  
double spin = gamepad1.left_stick_x * maxSpin;  
double s0 = tilt + spin;  
double s1 = tilt - spin;  
servo0.setPosition(s0);  
servo1.setPosition(s1);
```

Advanced Joystick Servo Control

Load Testing

After constructing the differential joints, I tested their ability to support and move weight. These tests helped determine whether the gear systems, bearings, and motors could handle the loads experienced by a full robotic arm.

The original servo-powered joint performed reliably and maintained smooth motion under moderate loads. The larger motor-driven base joint provided significantly greater torque and enabled the arm to support heavier weights.

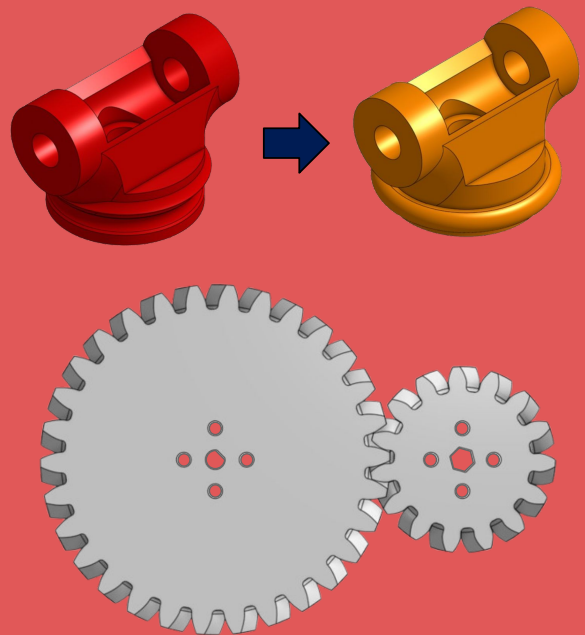
Tolerance Challenges

As the arm increased in size, small mechanical tolerances began to accumulate throughout the structure. This effect was most noticeable in the larger base joint, where the increased loads caused unwanted wobble and reduced positioning accuracy.

Design Improvements

To improve stability, I made several mechanical modifications:

- Replaced the ball bearing in the larger base joint with a sliding bearing
- Transitioned from triangular to trapezoidal bevel teeth to increase number of points of contact
- Fitted gear center holes directly to their respective axles to reduce reliance on axle hubs and set screws for maintaining alignment



Results

After implementing these improvements, the arm demonstrated smoother motion, improved stability, and greater reliability under load. The testing process helped identify weaknesses early and allowed the design to evolve into a more robust mechanical system.

Computer Vision Integration

After successfully designing and controlling the robotic arm using a gamepad, I expanded the project by integrating computer vision-based gesture control.



MediaPipe hand gesture model applied for coordinate tracking

Using the MediaPipe and OpenCV libraries, I developed a program capable of tracking hand movements in real time through a camera feed. Hand position data was processed and translated into control inputs for the robotic arm.

Gesture-Based Control

The goal was to mimic joystick movement using hand gestures.

The system:

- Tracked hand position relative to the camera frame
- Converted hand movement into virtual joystick inputs
- Allowed arm joints to follow hand motion in real time
- Maintained the same control logic previously developed for joystick operation

This approach allowed the robotic arm to be controlled without physical contact, demonstrating an innovative human-robot interaction method with significant potential for real-world application.

Final Results

- Multi-joint differential arm movement
- Reliable joystick-based control
- Stable operation under load
- Real-time gesture-based control using computer vision

The project evolved into a platform for exploring advanced mechanical design, innovative control systems, and human-robot interaction.



Final Project